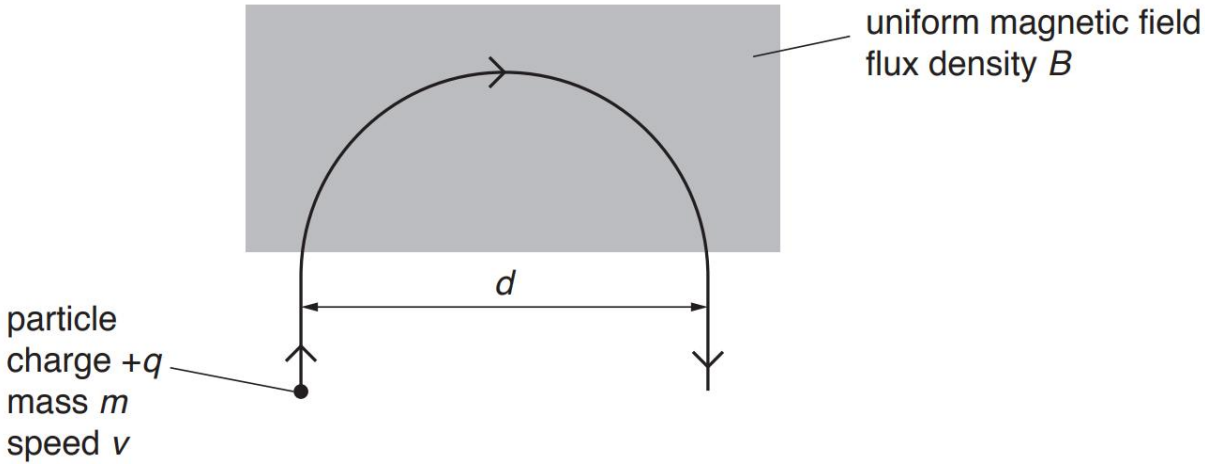
20.2.2 Exercises - Lorentz Force on a moving charge 2

1. A particle of charge +q and mass m is travelling in a vacuum with speed v. The particle enters, at a right angle, a uniform magnetic field of flux density B, as shown in Fig.



The particle leaves the field after following a semi-circular path of diameter d.

a. State the direction of the magnetic field.

..... [1]

b. Explain why the speed of the particle is not affected by the magnetic field.

.....

..... [2]

c. Show that the diameter d of the semi-circular path is given by the expression [2]

$$d = \frac{2mv}{Bq}.$$

d. Use the expression in (c) to show that the time T_F spent in the field by the particle is independent of its speed v. [2]

2.

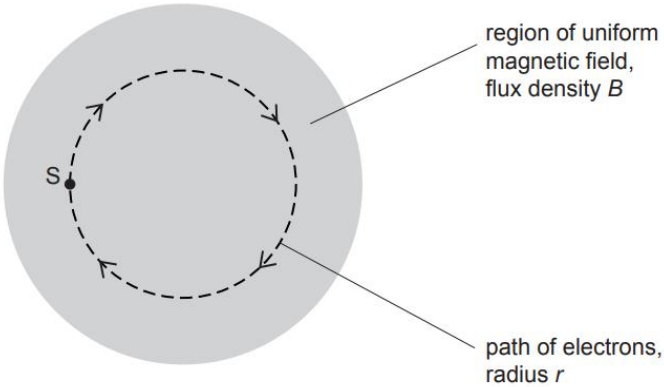
a. Define magnetic flux density.

.....

..... [2]

b. Electrons, each of mass m and charge q, are accelerated from rest in a vacuum through a potential difference V. Derive an expression, in terms of m, q and V, for the final speed v of the electrons. Explain your working. [2]

c. The accelerated electrons in (a) are injected at point S into a region of uniform magnetic field of flux density B, as illustrated in Fig.



The electrons move at right angles to the direction of the magnetic field. The path of the electrons is a circle of radius r.

i. Deduce the expression for the specific charge of the electrons [2]

ii. Electrons are accelerated through a potential difference V of 230 V. The electrons are injected normally into the magnetic field of flux density 0.38 mT.
The radius r of the circular orbit of the electrons is 14 cm.
Use this information to calculate a value for the specific charge of an electron.

specific charge = ... C kg^{-1} [2]

iii. Suggest why the arrangement outlined in (ii), using the same values of B and V, is not practical for the determination of the specific charge of α-particles.

.....

.....[2]

3.

a. State what is meant by a field of force.

.....

..... [2]

b. Explain the use of a uniform magnetic field and a uniform electric field for the selection of the velocity of charged particles. You may draw a diagram if you wish.

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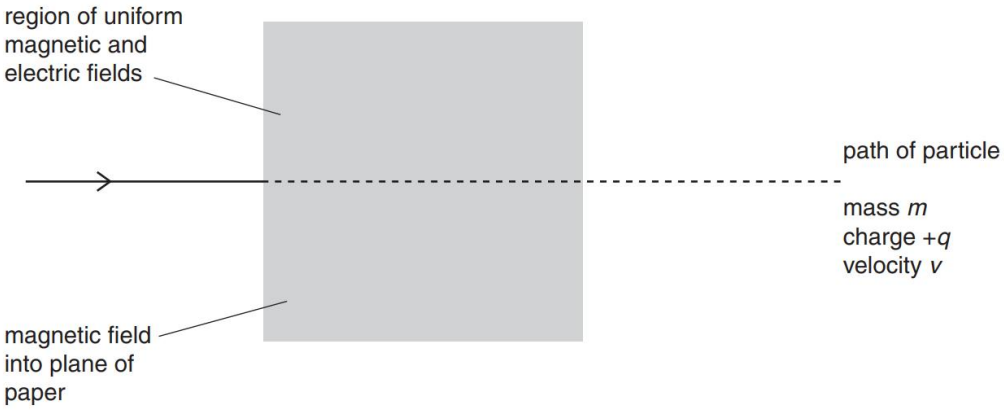
..... [4]

c. A beam of charged particles enters a region of uniform magnetic and electric fields, as illustrated in Fig.



The direction of the magnetic field is into the plane of the paper. The velocity of the charged particles is normal to the magnetic field as the particles enter the field.

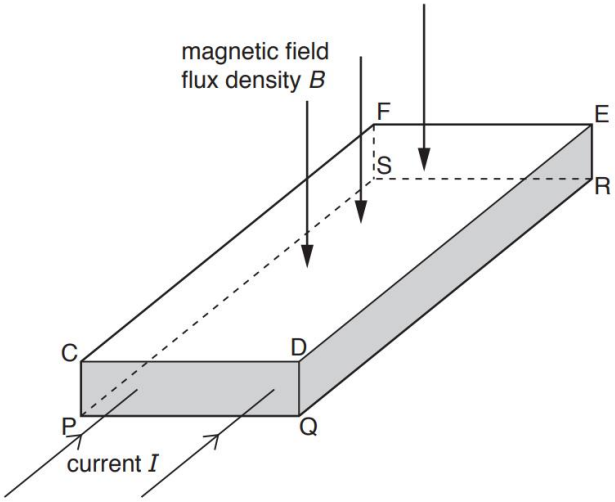
A particle in the beam has mass m, charge +q and velocity v. The particle passes undeviated through the region of the two fields.

On Fig., sketch the path of a particle that has

i. mass m, charge +2q and velocity v (label this path Q), [1]

ii. mass m, charge +q and velocity slightly larger than v (label this path V). [2]

4. A thin slice of conducting material is placed normal to a uniform magnetic field of flux density B, as shown in Fig.



The magnetic field is normal to face CDEF and to face PQRS.
A current I passes through the slice and is normal to the faces CDQP and FERS.
A potential difference, the Hall voltage VH, is developed across the slice.

a. State the faces between which the Hall voltage VH is developed.

..... and ..... [1]

b. The current I is produced by charge carriers, each of charge +q moving at speed v in the direction of the current. The number density of the charge carriers is n.

i. Derive an expression relating the Hall voltage VH to v, B and d, where d is one of the dimensions of the slice.[3]

ii. Use your answer in (b)(i) and an expression for the current I in the slice to derive the expression

$$V_H = \frac{BI}{ntq}.$$

Explain your working.[2]

c. Suggest why the Hall voltage is difficult to detect in a thin slice of copper.

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..... [2]